Project write up

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1 Introduction

Aims:
- Apply data science skills learnt in R to real data set (primary goal)
- Application of elementary statistical techniques to real life data sets
- Answer personal curiosities about distributions of public transport across England and Wales

The other month I read an opinion piece in my local village newsletter, written by a 90 year old woman who had forfeited her driving licence and had no options to continue going out of the village. She wrote about the adverse effects of losing the ability to drive in an area that is not serviced by any form of public transport including "How do I get to the doctors, get my hair done, my eyes tested and enjoy the fun things like shopping or meeting friends", "what I could not do is travel to any of these places by public transport"[1]. I am sure that is the first time the Titchmarsh Times has ever been cited. Being in a rural area it is my own anecdotal experience that connectivity by public transport is very sparse and hard to incorporate into travel for daily tasks. At this same time I had been reading articles about the utilisation of active transport in maintaining healthy active lifestyles [2] [4], and how the accessibility of active forms of transport influences how likely people were to choose active forms of transport. It is because of this I decided to investigate the distribution of public transport connectivity across rural and urban classified areas.

By definition, if an urban area is "a high density of residential addresses, or if they intersect with Amalgamated Built Up Areas (ABUAs) with a resident population of 10,000 or more" [5] then it would make sense that the overall connectivity through walking and cycling should be higher than rural areas, as these transport modes are more common over short distances. However, for public transport a relationship didn't seem so apparent. When combining the social benefits of public transport as well as the active component [7] it appears beneficial to have widespread access to public transport. Investigation of areas under-serviced could prove useful in planning new public transport routes.

Disclaimer: I am not attempting to make any causational claims but merely am eager to observe and quantify a suspicion that I already had about distribution of public transport connectivity over rural and urban areas. I am also aware that this is by no means a water tight methodology both in design and statistical analysis. All of this has been done without the aid of generative AI- which is very obvious based on how messy my R code is.

2 Methodology

The RUC (Rural-Urban Classification) data set from the ONS uses different levels to its classifications. First each LSOA (Lower layer Super Output Area) is designated as either "Urban" or "Rural", then further as " Urban", "Larger Rural" and "Smaller Rural". Initially the public transport connectivity on the highest level of classification was compared. This was done using an Welch Two Sample t-test. I am aware that family wise errors accumulate when performing the t-test more than once so I must continue learning about ANOVA as I know there could have been further statistical analysis on the 3 way classification.

$$H_0 : \mu_{\text{urban}} = \mu_{\text{rural}}$$

$$H_1 : \mu_{\text{urban}} - \mu_{\text{rural}} :\neq 0$$

Data concerning the connectivity at the Lower Layer Super Output Areas (LSOA's) was taken from the Department of Transport's connectivity metric, which covers the whole of England and Wales. The rurality data was taken from the ONS Rural Urban Classification (2021) [5]; however, the classification could not be used immediately for linear regression, as the classifications "larger rural", "Smaller Rural" and "Urban" being ordinal/categorical, do not have a numerical scale of rurality associated with them. A crude fix for this was to create a new variable that assigned an integer value of 1-3 based on the rural-urban classification, but this felt like too great an assumption to make and I had yet to sufficiently meet my goal of applying some basic data processing.

From this, a new numerical rating was created using the following equation to estimate the proportion of the population living in an urban environment for each LSOA by finding the proportion of each LSOA population living in an OA classified as urban.

$$\text{OA\_RUC}_i = \text{The Rural Urban classification for the ith OA}$$

$$\text{OA\_Population}_i = \text{The Population for the ith OA}$$

$$\text{Proportion of LSOA Urban} = \frac{\sum_{i=1}^{n} \mathbb{I}(\text{OA\_RUC}_i = \text{urban})\, \text{OA\_Population}_i}{\sum_{i=1}^{n} \text{OA\_Population}_i}$$

To create this data was accessed from the ONS's Census Output Area Population Estimates [6] and RUC on the OA level [5]. There is no distinction between "Larger Rural" and "Smaller Rural" in classification at the OA level unlike the

LSOA. Once this was created, linear regression of overall public transport connectivity score against the variables: "proportion of each LSOA population living in an OA classified as urban" and "Proportion of LSOA living near a major town ". The methodology used to classify living within 30 minutes of a major town can be found on the ONS RUC methodology [5].

3 Results

For the Welch Two Sample t-test comparing the means of "Urban" and "Rural" the t value of 169.22 over 7854 degrees of freedom gave an actual p value of $p < 2.2e\text{-}16$. The t value lies outside the 95 percent confidence interval of [32.3, 33.1] to 1 d.p. Therefore providing evidence to reject H_0. This was completed using R statistical software.

The linear model of the LSOA urban population yielded the following results figure 1:

Table 1

Results from the regression analysis

Predictor	t value	Pr(>|t|)	Gradient Estimate	Standard Error
Proportion of LSOA population living in an OA classified as urban	201.50	<2e-16	33.8190	0.1678
Proportion of LSOA living near to a major town	88.05	<2e-16	21.6777	0.2462

Figure 1: Results of the regression analysis

4 Discussion

Discussion: Both the t-test and the linear analysis present evidence of there being an unequal and statistically significant difference in the distribution of public transport access across rural-urban zones. Both tests have indicated that there is a higher amount of connectivity by public transport in increasingly urban zones. This could potentially have wider implications on potential health inequalities between urban and rural zones through unequal access to active transport.

The proportion of each LSOA living in an urban environment estimator uses a "first past the post" system and suffers bias based on how the OA are placed.

For example 51% of an OA could reside in a rural environment and therefore the OA is classified as rural, leaving the remainder of the OA that doesn't live in a rural environment excluded when calculating the proportion of the LSOA that reside in an urban environment. A solution to this would be to replicate the ONS's method of classification , but I believed that this was beyond this simple project and I lack the geospatial programming skills.

For this projects brevity I decided against including all of the individual public transport scoring in my analysis and only used the overall score given to public transport.

5 Personal Reflections

I have had great fun over the previous weeks learning about regression analysis, particularly linear regression. It has been fascinating to see how many different interpretations yield the same mathematical result. My personal favourite is the vector geometry of linear models by viewing the best coefficients as the coefficients of a vector projected onto the column space of the data matrix, feeling really intuitive and helpful for then finding degrees of freedom. This being said, understanding how to take partial derivatives of the cost function and optimisation through gradient descent has been very helpful to understand how logit and Poisson models are fitted. Then finding the unifying approach using the exponential family of distributions and link functions was very satisfying. I have completed a number of exercises from the book Introduction to Statistical Learning with R [3], as well as many other resources, and I am now keen to try and apply what I have learned to real life data. Being able to attempt to combine these new skills with projects on public health has been really interesting and a fun learning experience.

References

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[3] Gareth James et al. *An Introduction to Statistical Learning: With Applications in R.* Accessed 2026-05-04. Springer US, 2021.

[4] Anja Mizdrak et al. "Assessing the health impacts of changes in active transport: An updated systematic review". In: *Journal of Transport and Health* 33 (2024). URL: `https://www.sciencedirect.com/science/article/pii/S221414052300138X#bib9`.

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[6] Office for National Statistics. *Census Output Area Population Estimates (supporting information)*. Mar. 2024. URL: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/censusoutputareapopulationestimatessupportinginformation.

[7] Chris Rissel et al. "Physical Activity Associated with Public Transport Use—A Review and Modelling of Potential Benefits". In: *International Journal of Environmental Research and Public Health* 9.7 (July 2012). URL: https://www.mdpi.com/1660-4601/9/7/2454.